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| **Fiscal Year:** | FY 2020 | **Task Last Updated:** | FY 07/17/2020 |
| **PI Name:** | Masson, Patrick H. Ph.D. | | |
| **Project Title:** | Can Polyamines Mitigate Plant Stress Response under Microgravity Conditions? | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Bioregenerative Life Support | | |
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| **Zip Code:** | 53706-1580 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | 2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research |
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| **No. of Post Docs:** | | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | Zhang, Ye | **Contact Phone:** | 321-861-3253 |
| **Contact Email:** | Ye.Zhang-1@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | July 2020 report: Dr. Shih-Heng Su is an Assistant Scientist in the Masson lab who has been contributing to this project since its inception. | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | 80NSSC19K1483 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Space exploration requires the design and implementation of bioregenerative life-support systems capable of maintaining breathable air, cleaning water, and producing food and fiber during space missions. Plants are expected to play key roles in these life-support systems. Therefore, it is important to understand the effects the microgravity environment has on plants to allow the design and development of better life-support systems for space exploration. Multiple studies have demonstrated that plants grown under microgravity display signs of stress, including altered morphology and expression profiles expected for stress response. Therefore, the development of mitigation strategies aimed at alleviating such stress will be important. Previous studies by multiple research teams including ours have demonstrated a role for polyamines in stress mitigation in plants. Polyamines are a group of compounds whose abundance increases in plants under stress. These signaling molecules have also been shown to contribute to stress mitigation in ground-based experiments. Therefore, we hypothesize that polyamines might also contribute to stress mitigation under microgravity, and propose to test this hypothesis by investigating the growth and expression profiles of Arabidopsis thaliana seedlings engineered to modify the pool of putrescine (one of these polyamines) accumulating in their tissues. Specific objectives will include: 1) Comparing the growth of Arabidopsis thaliana seedlings with genetic alterations affecting the pool of putrescine to that of wild type, within the VEGGIE hardware, both under microgravity on the International Space Station (ISS) and on the ground (1-g control), and 2) Use RNAseq approaches to investigate the effect of microgravity relative to 1-g ground control on the transcriptomes of wild type and mutant seedlings with alterations in their putrescine pool. These investigations should lead to a better understanding of the roles played by polyamines in stress mitigation under microgravity. It may identify novel genetic engineering methods that improve plant growth under microgravity, thereby allowing development of optimized bioregenerative life-support systems for long-term space exploration missions. This project may also have important applications in agriculture for crop production on marginal lands. Finally, our experiments will yield large sets of transcriptomic data from previously uncharacterized genotypes, which will be added to the GeneLab dataset. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This project will lead to a better understanding of the role of Putrescine and/or its derived signaling products in mitigating the stress associated with growth under microgravity environments such as those encountered during spaceflight and on the International Space Station. It may also allow the identification of novel genetic engineering strategies aimed at improving plant adaptation to the microgravity environment, with potential impact on our ability to improve the genome of other plant species / crops for utilization in bioregenerative life-support systems during spaceflight. A better understanding of plant responses to stress may also have important applications in agriculture for crop production on marginal lands.<br>New large-scale RNAseq expression data associated with growth under microgravity for previously untested Arabidopsis genotypes will also be generated during this project and submitted to the GeneLab Data System.<br><br>This project will allow us to introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, and also generate novel instructional materials allowing effective teaching of the difficult concepts of gravity perception by plants and their importance for agricultural production on Earth and during spaceflight. Such materials should be usable in K-12, undergraduate, and graduate education, as well as general-public outreach. |
| | Plants grown under microgravity conditions display morphological and molecular phenotypes that are compatible with stress exposure. Interestingly, on Earth plants exposed to stress usually accumulate large amounts of polyamines, including putrescine and its derivatives spermidine and spermine, and these compounds typically contribute to stress mitigation [1]. However, analyses of plant transcriptional responses to microgravity suggest a failure to activate polyamine synthesis and response pathways. Considering these observations, we suggested that a failure for plants to activate polyamine synthesis and response pathways under microgravity conditions may be responsible, at least in part, for the altered growth and stress-like phenotypes displayed by microgravity-exposed plants. Advanced Plant Experiment (APEX)-06's main objective is to investigate the ability of wild type, mutant, and transgenic Arabidopsis plants with modified abilities to either accumulate or oxidize putrescine, to grow, develop and respond transcriptionally to the microgravity conditions of the International Space Station, and compare these growth and molecular responses to those of ground-control plants.<br>The following Arabidopsis thaliana genotypes will be used in these experiments: (1) wild type Col-0, (2) adc1-1 mutant affected in its ability to synthesize Put, (3) Cuao3-1 and cuao3-100cdr7 mutants affected in their ability to catabolize putrescine, (4) transgenic plants over-expressing the Poncirus trifoliata ADC (PtADC) gene (hereafter named PtADC-OX), and (5) transgenic cuao3-1 plants over-expressing the AtCuAO3 gene (hereafter named cuao3-1[CuAO3]-OX). Utilization of these genotypes should allow us to evaluate the effect of altering the pool of putrescine within Arabidopsis thaliana seedlings on plant responses to the stress encountered under microgravity, and also investigate a possible contribution of putrescine degradation products such as GABA and hydrogen peroxide, to plant stress mitigation under microgravity.<br><br>Last year, we generated some of the genetic material needed for these experiments, and also bulked up the corresponding seed stocks to generate enough material for the Science and Experimental Verification Tests (SVT and EVT) at Kennedy Space Center, and for the flight experiment and its ground control.<br><br>1) Seeds from wild-type Col-0, mutant cuao3-1, cuao3-100cdr7 and adc1-1, and transgenic cuao3-1[CuAO3]-OX, had previously been generated in the lab by former graduate student Amy Jancewicz [2]. We germinated and grew multiple plants for each one of these genotypes, and self-fertilized them, allowing the production of enough seeds from each genotype to carry out the SVT, EVT, flight experiment and its associated ground control. We also used PCR analysis to verify the genotypes of these plants. Furthermore, leaf material was collected from each plant and stored at -80°C for subsequent RNA extraction and expression studies. Current and future work is aimed at verifying the phenotype associated with each line, examining the level of targeted gene/transgene expression in these plants relative to wild type, and identifying independent transgenic lines that over-express the corresponding transgene relative to wild type.<br><br>2) We were also planning on using seeds from a transgenic Arabidopsis thaliana line that was previously shown to over-express the Poncirus trifoliata ADC (PtADC) gene, accumulate putrescine in their tissues, and display increased tolerance to abiotic stresses [3]. Unfortunately, we subsequently learned that this material had been lost by the lab that generated it. Therefore, we built a new construct aimed at over-expressing the PtADC transgene in Arabidopsis thaliana. A cDNA encoding the PtADC open reading frame was synthesized, fused to the CaMV 35S promoter and nos terminator sequences, and introduced into the pAGM4673 level-2 destination vector, along with a spectinomycin-resistance |

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| **Task Progress:** | selection cassette, using the Golden Gate technology [4]. A similar construct carrying an internally deleted version of the CaMV35Sp::PtADC-Nost cassette (with an internal 0.6 kbp BsaI segment deleted from the PtADC open-reading-frame) was also constructed for use as a negative control (hereafter named PtADC-OX). After verification by sequencing, these constructs were introduced into wild type Col-0, adc1-1 and adc2-1 mutant plants using the Agrobacterium tumefaciens-mediated floral-dip in planta transformation approach [5]. Transformed progeny were selected on spectinomycin-containing 0.8% agar-based 1/2MS media, propagated, and self-fertilized. Their progeny were again selected on spectinomycin-containing media, and propagated. Leaf samples were harvested from individual transformants, and frozen in liquid nitrogen. RNA is being extracted from these samples, which will be used to evaluate transgene expression levels in independent transgenic lines, using RT-qPCR approaches. Interestingly, some of these PtADC-OX transgenic plants displayed distinct levels of growth and morphological phenotypes depending on the line analyzed, including shorter organs, spiraling leaves, and partial to complete sterility. These phenotypes were not observed on plants transformed with the internally deleted version of PtADC (PtADC-OX). The variable nature of these phenotypes between independently isolated lines suggests that they might derive from variations in levels of transgene over-expression due to position effects. We are testing this hypothesis by investigating the level of transgene expression in leaf tissues from these plants relative to wild type, using RT-qPCR approaches. Growth and morphological phenotypes will be compared between lines showing distinct levels of transgene over-expression. Additional experiments will also allow evaluation of plant sensitivity to abiotic stresses such as osmotic, salt, drought, and hypoxia.<br><br>We expect higher levels of transgene expression in plants displaying the strongest phenotypes, and lower levels in normal looking plants [6]. We also expect increased tolerance to at least some of these stresses for over-expressing transgenic plants relative to wild type, and increased sensitivity to stress for the corresponding knockout mutant plants.<br><br>These experiments will allow us to select transgenic plants that appear morphologically normal, but still display increased transgene expression levels relative to wild type, along with altered tolerance to abiotic stresses, for use in our SVT, EVT, and spaceflight experiments.<br><br>This project was spearheaded by Dr. Shih-Heng Su, helped by undergraduate student Tyler Serie Anne Ganser for plant cultivation and seed bulking. The Covid-19 lockdown interfered with progress in this project, starting mid-March 2020 and lasting until the end of June, 2020. However, we were still allowed to continue bulking seeds from our wild type, mutant and transgenic plants during this period, a process that allowed us to generate enough material to test our lines and, in the near future, carry out the SVT, EVT, flight and ground control experiments. Plant tissues were also harvested during this period, which we are currently using to investigate target gene expression levels in our lines.<br><br>Cited Literature<br><br>[1] Jancewicz, A., Gibbs, N. & Masson, P. Cadaverine's functional role in plant development and environmental response. Front Plant Sci 7, 1-8, doi:10.3389/fpls.2016.00870 (2016).<br><br>[2] Jancewicz, A. A Genetic Study of Cadaverine Response Reveals Crosstalk Between Cadaverine and Putrescine Pathways in Arabidopsis thaliana PhD thesis, University of Wisconsin-Madison, (2017).<br><br>[3] Wang, J. et al. An arginine decarboxylase gene PtADC from Poncirus trifoliata confers abiotic stress tolerance and promotes primary root growth in Arabidopsis. J Exp Bot 62, 2899-2914, doi:10.1093/jxb/erq463 (2011).<br><br>[4] Engler, C. et al. A golden gate modular cloning toolbox for plants. ACS Synth Biol 3, 839-843, doi:10.1021/sb4001504 (2014).<br><br>[5] Bent, A. Arabidopsis in planta transformation. Uses, mechanisms, and prospects for transformation of other species. Plant Physiol 124, 1540-1547 (2000).<br><br>[6] Alcázar, R., García-Martinez, J., Cuevas, J., Tiburcio, A. & Altabella, T. Overexpression of ADC2 in Arabidopsis induces dwarfism and late-flowering through GA deficiency. Plant J 43, 425-436, doi:10.1111/j.1365-313X.2005.02465.x (2005). |
| **Bibliography Type:** | Description: (Last Updated: 07/03/2024) |
| **Articles in Peer-reviewed Journals** | Su S-H, Keith MA, Masson PH. "Gravity signaling in flowering plant roots." Plants. 2020 Oct;9(10):1290. [Published online: 29 September 2020.] https://doi.org/10.3390/plants9101290 ; PMID: 33003550; PMCID: PMC7601833 , Oct-2020 |